

PulmoScan: An AI-Based Hybrid Approach for Early Detection and Classification of Lung Nodules Using Deep Learning on CT Scans

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Abstract—Lung cancer, a leading cause of global mortality, often goes undetected until late stages. PulmoScan introduces an AI-based hybrid approach for early detection and classification of lung nodules using CT scans. Employing YOLOv8, Faster R-CNN, EfficientNet, DenseNet, MobileNetV2, and ResNet18, the system automates nodule detection and classification, reducing radiologist workload and enabling timely treatment.

Index Terms—Lung cancer detection, Pulmonary nodules, YOLOv8, EfficientNet, Deep learning, Medical imaging, AI in healthcare

I. INTRODUCTION

Lung cancer remains a global health crisis, claiming 1.8 million lives annually and ranking as one of the leading causes of cancer-related deaths [1]. Its high mortality is largely due to late-stage diagnoses, as symptoms often remain silent until the disease has progressed significantly. Early detection can dramatically improve survival rates by up to 40%, yet current diagnostic processes are slow, often taking 3 to 6 months due to reliance on manual radiologist reviews [2]. These reviews are not only time-consuming but also prone to human error, particularly under the strain of increasing patient volumes and limited specialist availability. In resource-constrained settings, these challenges are amplified, leading to delays that hinder timely interventions. PulmoScan addresses these issues by leveraging artificial intelligence to automate the detection and classification of lung nodules on CT scans. By integrating advanced deep learning models, PulmoScan aims to enhance diagnostic accuracy, reduce radiologist workload, and accelerate the path to treatment, ultimately improving patient outcomes worldwide.

II. RELATED WORK

Recent AI advancements have improved lung nodule detection. Similar projects in Tunisia, such as those at the University of Sfax, have developed AI tools for medical imaging [4]. In Africa, initiatives like those in South Africa have used AI for chest disease detection [5]. Additional efforts in Tunisia include work at the University of Monastir, where researchers applied deep learning to detect pulmonary abnormalities in CT scans [6]. Across Africa, a Nigerian team developed an AI system for tuberculosis detection using chest imaging, demonstrating applicability in low-resource settings [7]. Globally, projects in India have explored CNN-based lung cancer

detection [8], while in Brazil, researchers implemented AI for nodule classification [9]. PulmoScan builds on these efforts, offering a hybrid pipeline for both detection and classification, tailored for clinical use.

III. METHODOLOGY

We adopted the Team Data Science Process (TDSP), a structured methodology that ensures systematic project execution, enhances collaboration, and aligns AI development with clinical needs, making it ideal for our multidisciplinary team.

A. Data Acquisition and Understanding

1) Data Acquisition:

- **LUNA16**: 888 CT scans
- **TCIA**: 1,200 CT scans
- **IQ-OTHNCCD**: 1,097 CT scans

2) Data Preprocessing:

- **LUNA16**: Gaussian/Poisson noise
- **TCIA**: Motion blur, speckle noise
- **OTHNCCD**: Gaussian blur, contrast
- **All**: Normalization, rotation, flipping

B. LUNA16 Dataset Visualization

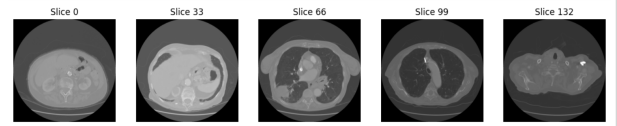


Fig. 1: Preprocessed CT slice from LUNA16 dataset showing Gaussian noise effect.

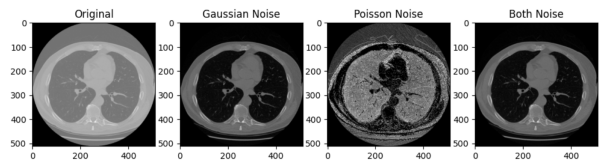


Fig. 2: CT slices from patient LIDC-IDRI-0166 in the LUNA16 dataset: (a) Slice 0; (b) Slice 33; (c) Slice 60; (d) Slice 99; (e) Slice 132.

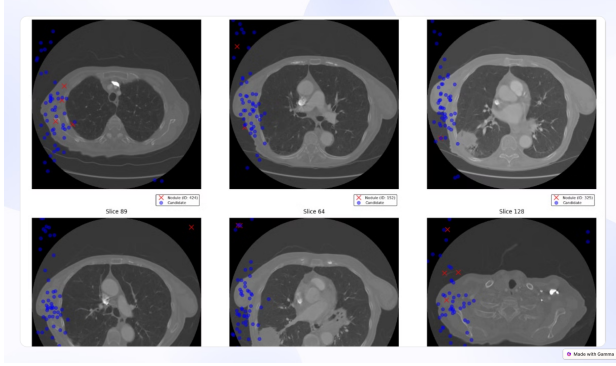


Fig. 3: Nodule annotations on LUNA16 dataset: Blue dots indicate candidate nodules; red X's mark confirmed nodules.

C. TCIA Dataset Visualization

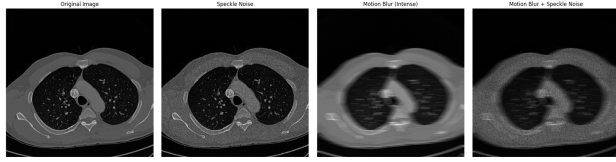


Fig. 4: Original CT scan from a representative patient (TCIA).

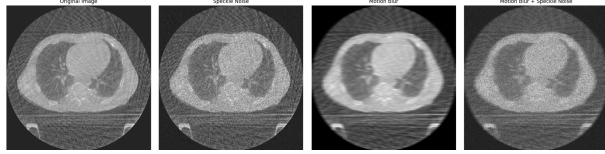


Fig. 5: CT scan with motion blur to simulate patient movement (TCIA).

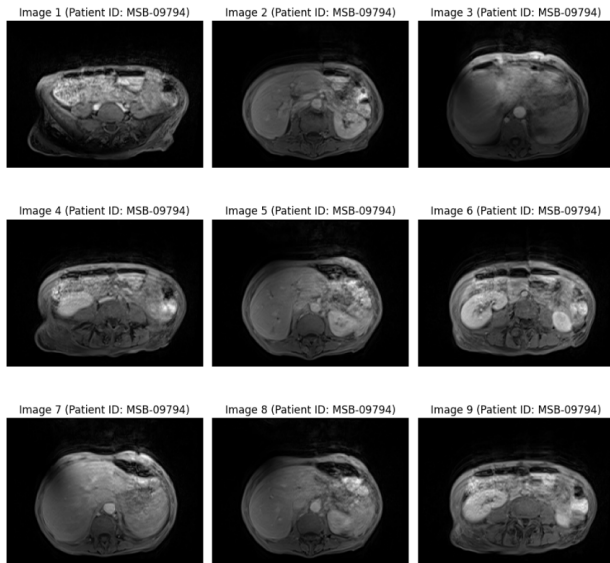


Fig. 6: CT slices from TCIA scan: (a) Slice 10; (b) Slice 30; (c) Slice 60; (d) Slice 90; (e) Slice 120.

D. IQ-OTHNCCD Dataset Visualization

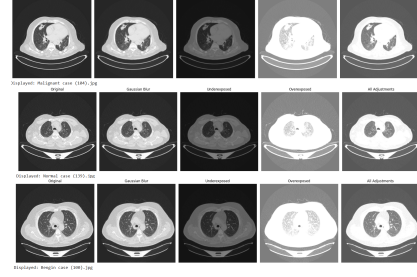


Fig. 7: IQ-OTHNCCD dataset: (a) Original CT scan; (b) Gaussian blur; (c) Contrast variation.

E. Model Architecture

- **Nodule Detection:** YOLOv8Fast was used as a lightweight and efficient model for detecting lung nodules on CT scans.
- **Primary Classification:** MobileNetV2 and DenseNet121 were used to classify nodules as benign or malignant.
- **Secondary Classification:** ResNet18, DenseNet121, EfficientNetB0, and Xception were employed to predict the subtype of non-small cell lung cancer.
- **Chosen Models for Deployment:**
 - **Primary Classification:** MobileNetV2
 - **Secondary Classification:** ResNet18
 - **Detection:** YOLOv8Fast

IV. EXPERIMENTS AND RESULTS

The models used include YOLOv8Fast for detection, MobileNetV2 for primary classification (benign vs. malignant), and ResNet18 for secondary classification (NSCLC subtypes). Their definitions are provided in Table I.

TABLE I: Definitions of Models Used in PulmoScan

Model	Definition
YOLOv8Fast	A lightweight object detection model optimized for speed and accuracy in detecting lung nodules.
MobileNetV2	A lightweight CNN designed for efficient classification of nodules as benign or malignant.
ResNet18	A deep residual network balancing accuracy and speed for classifying NSCLC subtypes.

We conducted experiments using preprocessed CT images from LUNA16, TCIA, and IQ-OTHNCCD datasets. Each model was trained and validated using stratified k-fold cross-validation (k=5). Performance was evaluated based on accuracy, precision, recall, F1-score, and ROC-AUC.

A. Nodule Detection

YOLOv8 achieved a detection accuracy of 92.4% on the TCIA dataset with high speed, suitable for real-time clinical settings.

B. Primary Classification (Benign vs. Malignant)

- **MobileNetV2:** Accuracy: 96%, F1-score: 0.96
- **DenseNet121:** Accuracy: 95%, F1-score: 0.95

C. Secondary Classification (NSCLC Subtypes)

- **ResNet18:** Accuracy: 97.23%
- **DenseNet121:** Accuracy: 89.84%
- **EfficientNetB0:** Accuracy: 89%
- **Xception:** Accuracy: 82%

ResNet18 was chosen for deployment due to its balance between accuracy and speed.

V. DISCUSSION

PulmoScan's hybrid pipeline, using YOLOv8Fast, MobileNetV2, and ResNet18, enables efficient deployment in low-resource settings. Preprocessing enhanced robustness against image noise, addressing limitations of prior approaches. Future integration with hospital PACS systems can streamline workflows.

VI. CONCLUSION AND FUTURE WORK

PulmoScan offers an AI-based solution for early lung nodule detection and classification. Future work includes:

- Extending to other thoracic diseases.
- Incorporating 3D CNNs for volumetric analysis.
- Validating with clinical trials.
- Deploying a web-based interface.
- Exploring the NIH Chest X-ray dataset for additional insights.

ACKNOWLEDGMENTS

We thank Esprit Honoris United Universities and our Tutors for their support.

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APPENDIX

Code and trained models will be made available on GitHub upon publication.